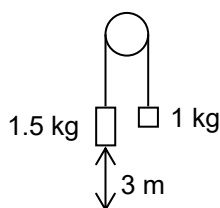


Pulleys Introduction

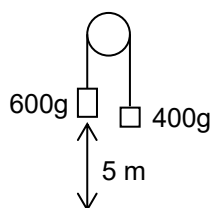
I.

The following systems are released from rest. For each, find the acceleration of the parcels, decide which will hit the ground and after what time. With what speed does the parcel hit the ground? All pulleys are smooth and freely turning; the strings are light and inextensible and the air resistance negligible.

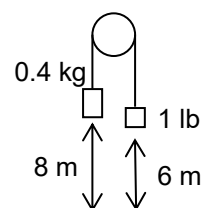
1.



2.



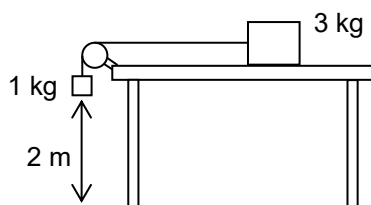
3.



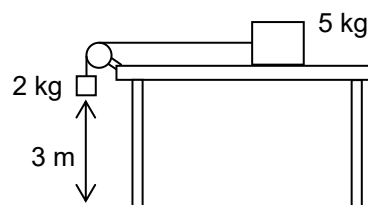
II.

Find the time taken for the following parcels to hit the ground, after the systems are released from rest. Also find the tension in the string. Assume that the surface is smooth and horizontal.

4.



5.



III.

Three weights, of masses 7 kg, 10 kg and 4 kg, are connected by light inextensible *strings* running over smooth pulleys as shown in the diagram. The 10 kg mass lies on a smooth horizontal table and the other two masses hang freely. Find the acceleration of the objects and hence find the tension in both strings.

